

Past the critical dimension: the fractional filter for $d \geq 4$

Working note for the “Keep the Angle” program

Purpose. The rank form of Angular Preservation is settled below the critical dimension $d = 4$: the commute angular ranking converges uniformly in the mode count to the Green-kernel ranking (Paper I, Green-kernel rank-limit theorem), reducing the question to Green-rank positivity. At $d = 4$ the commute Green kernel $G = \sum_k \mu_k^{-1} \varphi_k \otimes \varphi_k$ ceases to be Hilbert–Schmidt ($\sum_k \mu_k^{-2} = \infty$ by Weyl’s law $\mu_k \asymp k^{2/d}$), so the limit step fails. Paper I flags the remedy—a stronger filter $\mu_k^{-s/2}$, $s > d/4$ —and the first question, monotonicity of the fractional Green kernel on two-point homogeneous spaces. This note carries the entire chain up one level to the fractional filter and, using heat-kernel subordination, proves the two-point homogeneous case *in every dimension*, so that the reduction “Angular Preservation $\Leftarrow$ fractional-Green-rank positivity” holds for all d , with the geometric coefficient equal to 1 on the rank-one symmetric spaces regardless of dimension. Notation follows Paper I, the Green-kernel rank-limit note, and the companion note “green-rank-positivity”.

1 The fractional filter and Hilbert–Schmidt stability

Fix $s > 0$ and, at a multiplet-closed cutoff Λ , define the s -filtered embedding and its kernel

$$F_{s,\Lambda} = (\mu_k^{-s/2} \varphi_k)_{0 < \mu_k \leq \Lambda}, \quad K_{s,\Lambda}(x, y) = \sum_{0 < \mu_k \leq \Lambda} \frac{\varphi_k(x) \varphi_k(y)}{\mu_k^s}, \quad S_{s,\Lambda}(x) = K_{s,\Lambda}(x, x), \quad N_{s,\Lambda} = \frac{F_{s,\Lambda}}{\sqrt{S_{s,\Lambda}}}.$$

The commute filter is $s = 1$. Write $K_s = \sum_{k \geq 1} \mu_k^{-s} \varphi_k \otimes \varphi_k$ for the full s -kernel (the kernel of Δ^{-s} on the zero-mean subspace) and $\rho_{G,s}(M) := \rho_S(d_M(X, Y), -K_s(X, Y))$ for the *fractional Green-rank coefficient*.

Lemma 1 (Hilbert–Schmidt stability). $\|K_{s,\Lambda'} - K_{s,\Lambda}\|_{L^2(M \times M)}^2 = \sum_{\Lambda < \mu_k \leq \Lambda'} \mu_k^{-2s}$, and by Weyl’s law $\mu_k \asymp k^{2/d}$ this converges iff $s > d/4$. Hence for $s > d/4$, $K_{s,\Lambda} \rightarrow K_s$ in $L^2(M \times M)$, and $K_s \in L^2$.

Proof. The $\varphi_k \otimes \varphi_k$ are orthonormal in $L^2(M \times M)$, giving the stated norm; $\sum_k \mu_k^{-2s} \asymp \sum_k k^{-4s/d}$ converges iff $4s/d > 1$. $\square$

For $d \geq 4$ any $s > d/4 > 1$ works; the minimal admissible filter grows with dimension.

2 The fractional rank limit

Theorem 2 (Fractional Green-kernel rank limit). *Let M be closed of dimension d , $\text{vol}(M) = 1$, and $s > d/4$. Let (X, Y) have bounded density with respect to $\text{vol} \otimes \text{vol}$, with the laws of $d_M(X, Y)$ and $K_s(X, Y)$ atomless. Then*

$$\rho_S(d_M(X, Y), \|N_{s,\Lambda}(X) - N_{s,\Lambda}(Y)\|) \longrightarrow \rho_S(d_M(X, Y), -K_s(X, Y)) = \rho_{G,s}(M),$$

and likewise for Kendall's τ . Consequently, if $\rho_{G,s}(M) > 0$ there are $\Lambda_0 < \infty, \rho_0 > 0$ with the s -filtered angular ranking $\geq \rho_0$ for every multiplet-closed $\Lambda \geq \Lambda_0$.

Proof. Identical in structure to the $s = 1$ theorem (Paper I, Green-kernel rank-limit), with μ_k^{-1} replaced by μ_k^{-s} throughout. (1) Hilbert–Schmidt limit is Lemma 1. (2) The diagonal $S_{s,\Lambda}(x) = \sum_{\mu_k \leq \Lambda} \mu_k^{-s} \varphi_k(x)^2$ is asymptotically constant uniformly in x : the uniform pointwise Weyl law $\sum_{\mu_k \leq L} \varphi_k(x)^2 = c_d L^{d/2} + O(L^{(d-1)/2})$ (Hörmander, uniform in x) and partial summation give, for $d/4 < s \leq d/2$, $S_{s,\Lambda}(x) = A_{s,\Lambda} + o(A_{s,\Lambda})$ uniformly in x , where $A_{s,\Lambda}$ is the leading term of $\int^\Lambda L^{-s} dc_d L^{d/2} \asymp \Lambda^{d/2-s}$ (diverges) for $s < d/2$, $\asymp \log \Lambda$ for $s = d/2$ (the uniform Hörmander remainder $O(L^{(d-1)/2})$ contributes a relative $O(\Lambda^{-1/2})$, lower order). For $s > d/2$ the diagonal instead converges to a finite, generally non-constant $S_s(x)$ and does not rescale to a constant—this case is the Remark. (3) Rescale $H_{s,\Lambda} = A_{s,\Lambda} C_{s,\Lambda} \rightarrow K_s$ in L^2 , ranks unchanged by the positive constant. (4) Bounded pair density carries L^2 -convergence to convergence in probability; atomlessness of the limit gives distribution-function convergence and hence convergence of the population Spearman/Kendall coefficients. $\square$

Remark 3 (The convergent-diagonal regime $s > d/2$). When $s > d/2$ the diagonal $S_{s,\Lambda}(x) \rightarrow S_s(x) = \sum_k \mu_k^{-s} \varphi_k(x)^2$, a finite, generally non-constant function of x (the fractional analogue of a local density). Step (2) then does not rescale to a constant; instead $C_{s,\Lambda} \rightarrow K_s(x, y) / \sqrt{S_s(x) S_s(y)}$ pointwise and in L^2 , and the rank limit is $\rho_S(d_M, -K_s / \sqrt{S_s S_s})$. The normalization by S_s is exactly the row-normalization of the s -embedding; on a homogeneous space S_s is constant and this reduces to $\rho_{G,s}$. For the inhomogeneous case the operative coefficient is the S_s -normalized fractional Green-rank correlation, and Theorem 2 holds with K_s replaced by $K_s / \sqrt{S_s(x) S_s(y)}$; all subsequent statements carry the same replacement. We keep K_s in the display for the homogeneous and $d/4 < s \leq d/2$ cases, where $S_{s,\Lambda}$ genuinely rescales to a constant.

3 The fractional master inequality and the Riesz parametrix

The master inequality is filter-agnostic: it used only that $-K$ is compared to a monotone radial profile. Restating (companion note, “green-rank-positivity”, Theorem “Master inequality”): for any strictly increasing Ψ with remainder $R_s = -K_s - \Psi \circ d_M$,

$$\rho_{G,s}(M) \geq 1 - 3 \mu_\Psi(2 \|R_s\|_\infty) \geq 1 - 12 \phi_\Psi \|R_s\|_\infty,$$

with $\rho_{G,s} = 1$ when $R_s \equiv 0$. The near-diagonal remainder is again bounded, now by the Riesz parametrix.

Lemma 4 (Riesz parametrix). *Let M be closed smooth, $\text{vol}(M) = 1$, and $s > (d-1)/2$ (a nonempty range for every d , and automatically $> d/4$ since $d > 2$). Near the diagonal $K_s(x, y) = \gamma_{d,s} d_M(x, y)^{2s-d} + h_s(x, y)$ with $\gamma_{d,s} > 0$ the universal Riesz constant and h_s bounded up to the diagonal; the leading profile $r \mapsto \gamma_{d,s} r^{2s-d}$ (exponent $2s-d \in (-1, 0)$, strictly decreasing) is radial with a point-independent coefficient, and it is the only singular part—the subleading Hadamard corrections carry exponent $2s-d+2j \geq 2s-d+2 > 1$ for $j \geq 1$, hence are bounded even though their coefficients are point-dependent. When $2s \geq d$, K_s is bounded (continuous, $2s > d$) or $-c \log d_M + \text{bounded}$ ($2s = d$). In all cases $\Psi_s = -\gamma_{d,s} r^{2s-d}$ (resp. the bounded/log profile), extended strictly increasing on $[r_0, \text{diam } M]$, has finite $\|R_s\|_\infty$ and finite ϕ_{Ψ_s} .*

Proof. $K_s = \Delta^{-s}$ on the zero-mean subspace is the Riesz potential of order $2s$; it is a classical pseudodifferential operator of order $-2s$ with principal symbol $|\xi|_g^{-2s}$, and its kernel has the Hadamard expansion $K_s(x, y) = \sum_{j \geq 0} c_j(x, y) d_M(x, y)^{2s-d+2j} + (\text{smooth})$ with $c_0 = \gamma_{d,s}$ the universal Euclidean Riesz constant (the metric is Euclidean to second order in normal coordinates) and c_j smooth for $j \geq 1$. The restriction $s > (d-1)/2$ makes $2s-d > -1$, so every $j \geq 1$ term has exponent $2s-d+2j > 1$ and is bounded up to the diagonal (with bounded, point-dependent coefficient); thus only $c_0 d_M^{2s-d}$ is singular, and $h_s := K_s - \gamma_{d,s} d_M^{2s-d}$ is bounded. Hence $R_s = -K_s - \Psi_s \circ d_M = -h_s$ near the diagonal, bounded; off the diagonal both K_s and $\Psi_s \circ d_M$ are bounded. Finiteness of ϕ_{Ψ_s} is the change-of-variables computation of the companion note with exponent $2s-d$ in place of $2-d$: $\Psi'_s \asymp r^{2s-d-1}$ and $f_W \asymp r^{d-1}$ give density $\asymp r^{2(d-s)}$, which $\rightarrow 0$ at $r \rightarrow 0$ for $s < d$, and is bounded on $[r_0, \text{diam } M]$. $\square$

Theorem 5 (Fractional near-radial positivity). *Let $s > (d-1)/2$. With Ψ_s the Riesz profile of Lemma 4 (a single fixed profile, both ϕ_{Ψ_s} and $\|R_s\|_\infty$ finite), $\rho_{G,s}(M) \geq 1 - 12\phi_{\Psi_s}\|R_s\|_\infty$, so $\rho_{G,s}(M) > 0$ whenever $\phi_{\Psi_s}\|R_s\|_\infty < 1/12$; and by Theorem 2 the s -filtered angular ranking is then uniform-in-mode-count rank-faithful. The perturbative openness argument (companion reductions note) transfers with $\mu_k^{-1} \rightarrow \mu_k^{-s}$ —its spectral-tail step now requires $\sum_k \mu_k^{-2s} < \infty$ (i.e. $s > d/4$), the resolvent contour weight becomes z^{-s} (holomorphic off the negative axis, with 0 excluded), and atomlessness of the K_s -pair law is assumed as in Theorem 2—so $\rho_{G,s} > 0$ on a C^2 -open set of metrics containing the exact cases of §4.*

4 Two-point homogeneous spaces, all dimensions, via subordination

The one place the fractional story is *cleaner* than the commute case is the exact computation: there is no local Laplace ODE for Δ^{-s} , but subordination to the heat semigroup supplies strict monotonicity in every dimension at once.

Theorem 6 (Fractional Green-rank = 1 on rank-one symmetric spaces). *Let M be a compact two-point homogeneous space (a sphere S^d , or a projective space $\mathbb{RP}^d, \mathbb{CP}^d, \mathbb{HP}^d, \mathbb{OP}^2$) of any dimension, $\text{vol}(M) = 1$, and let $s > 0$. Then $K_s(x, y) = k_s(d_M(x, y))$ with k_s strictly decreasing on $(0, \text{diam } M)$; hence $\rho_{G,s}(M) = 1$, and the S_s -normalized coefficient of the Remark is likewise 1. In particular, for every d and every admissible $s > d/4$, Angular Preservation (fractional-filter rank form) holds on these spaces, uniformly in the mode count.*

Proof. By subordination, for $s > 0$ and $\lambda > 0$, $\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t} dt$; applied to Δ on the zero-mean subspace,

$$K_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} (p_t(x, y) - 1) dt,$$

where p_t is the heat kernel and the -1 removes the constant mode ($\text{vol } M = 1$). Two-point homogeneity makes p_t a function of geodesic distance, $p_t(x, y) = P_t(d_M(x, y))$, and on a compact two-point homogeneous space P_t is *strictly decreasing* in the distance for every $t > 0$. This is the one genuinely analytic input; we justify it by the parabolic maximum principle rather than by citation. Writing the radial heat equation $\partial_t P = \partial_r^2 P + m(r) \partial_r P$ with $m = A'/A$ the log-derivative of the geodesic-sphere area density—on a rank-one symmetric space $m(r) = \sum(\text{mult}) \cdot \frac{\alpha}{2} \cot \frac{\alpha r}{2}$, strictly decreasing—the radial derivative $v := \partial_r P_t$ solves $\partial_t v = \partial_r^2 v + m \partial_r v + m'(r) v$ with $v = 0$ at the pole and the antipodal focal set and $m'(r) < 0$. Since $v < 0$ for small t (the

kernel is a decreasing peak) and the lower-order coefficient m' is negative, the strong parabolic maximum principle keeps $v < 0$ on the open interval $(0, \text{diam } M)$ for all $t > 0$ (individual zonal harmonics oscillate, but the heat-weighted sum does not). Differentiating K_s under the integral, which is justified because $\partial_r P_t(r)$ is integrable against t^{s-1} (small- t Gaussian decay $\partial_r P_t = O(t^{-d/2-1} r e^{-r^2/4t})$ times t^{s-1} is integrable at 0 for $r > 0$; large- t exponential relaxation $|\partial_r P_t| \leq C e^{-\mu_1 t}$ times t^{s-1} is integrable at ∞),

$$k'_s(r) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \partial_r P_t(r) dt < 0 \quad (0 < r < \text{diam } M),$$

since $t^{s-1} > 0$ and $\partial_r P_t(r) < 0$ for every $t > 0$. Thus $-K_s$ is a strictly increasing function of d_M , the pair $(d_M, -K_s)$ is comonotone, and $\rho_{G,s} = 1$. Normalizing by the constant diagonal S_s (homogeneous space) does not change ranks. $\square$

Remark 7 (What this settles). Theorem 6 removes the dimension ceiling from the exact case: the fractional Green-rank coefficient is 1 on all rank-one symmetric spaces in *every* dimension, for every $s > 0$ —strictly stronger than the $s = 1$, $d \leq 3$ corollary of Paper I, and proved by a different (subordination) route that never needs Hilbert–Schmidt stability. Combined with Theorem 2 ($s > d/4$) and Theorem 5 (openness), the reduction of Angular Preservation to Green-rank positivity now holds in all dimensions with the fractional filter, and its exact anchor—value 1 on the symmetric spaces—is dimension-free. The residual open problem is the same shape as before, one level up: fractional-Green-rank positivity $\rho_{G,s}(M) > 0$ for general M , with the near-radial class of Theorem 5 as the proven interior.

5 Imported facts and status

Imported. (1) Weyl’s law $\mu_k \asymp k^{2/d}$ and the uniform pointwise Weyl law with remainder $O(L^{(d-1)/2})$ (Hörmander). (2) Riesz-potential/heat-parametrix structure of Δ^{-s} : classical Ψ DO of order $-2s$, kernel singularity $\gamma_{d,s} d_M^{2s-d}$ with universal leading constant (Seeley; Gilkey, *Invariance Theory*). (3) Subordination formula $\lambda^{-s} = \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-\lambda t} dt$. (4) Strict monotonicity in distance of the heat kernel on compact two-point homogeneous spaces, via the parabolic maximum principle applied to $v = \partial_r p_t$ (proved in Theorem 6; classical on S^1 through the theta function). (5) The master inequality and openness machinery (companion notes), filter-agnostic.

Status. For every d and $s > d/4$: the fractional angular ranking converges uniformly in the mode count to the fractional Green-kernel ranking (Theorem 2); its coefficient is 1 on all rank-one symmetric spaces (Theorem 6, all $s > 0$) and, for $s > (d-1)/2$, > 0 on an explicit near-radial / C^2 -open class (Theorem 5). *Open*: fractional-Green-rank positivity for general metrics past the near-radial class (the $d \geq 4$ analogue of the standing conjecture), and an effective lower bound on the admissible s beyond $s > d/4$ if a sharper diagonal control is wanted.